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AI-Ready Payments: Why Legacy Payment Architectures Cannot Support Safe, Explainable, and Real-Time

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The real-time transparency problem in payments

Payments operate at a fundamentally different speed and risk profile than other banking functions. While most legacy cores were designed for batch processing, payments demand **millisecond-level decisioning**, real-time fraud detection, and immediate exception handling.

AI promises to enhance routing, reduce fraud, and improve straight-through processing (STP), but it cannot function safely when the underlying payment architecture is opaque. Legacy payment systems—built on COBOL routines, hard-coded rules, and undocumented patches—cannot provide the transparency AI requires to justify or explain decisions.

This creates a structural mismatch: **AI is real-time; legacy payments are not.**

The result is a transparency gap that exposes banks to operational, financial, and regulatory risk.

Hidden dependencies inside legacy payment flows

Payment systems accumulate complexity over decades. Beneath the surface of every ACH, wire, SWIFT, or RTP transaction lies a web of hidden dependencies:

- Routing logic embedded in monolithic COBOL modules
- Exception-handling rules that evolved through emergency patches
- Fraud scoring dependent on legacy data paths
- Downstream systems influenced by undocumented interfaces
- ISO 20022 transformations layered on top of older message formats

These dependencies create a **black-box payment architecture** where neither humans nor machines can trace how a decision was formed.

AI models trained on such environments inherit these blind spots. Even if the model is explainable, the system it interacts with is not—making the entire payment decision chain opaque.

Why AI fails in legacy payment systems

AI in payments requires:

- Clean, real-time data
- Deterministic routing logic
- Traceable decision paths
- Consistent lineage
- High-fidelity observability

Legacy payment systems provide none of these. Instead, they introduce:



- Latency from batch-oriented cores
- Inconsistent data lineage
- Hard-coded rules that override AI decisions
- Opaque exception flows
- Limited auditability

This leads to unpredictable outcomes, failed STP, and regulatory exposure.

The issue is not the AI model—it is the architecture beneath it.

The Payment Transparency Maturity Model (PTMM)

To address these challenges, payment modernization requires a structured approach. The **Payment Transparency Maturity Model (PTMM)** provides a five-level framework for assessing and improving payment architecture readiness for AI.

Level 1 — Opaque Payment Logic

Routing, fraud rules, and exception handling are embedded in legacy code with no documentation or traceability.

Level 2 — Partial Lineage Visibility

Some data flows are mapped, but dependencies remain hidden across channels and systems.

Level 3 — Component-Level Transparency

Business logic is decoupled into modular components with traceable decision paths.

Level 4 — Real-Time Observability

Payment flows, routing decisions, and fraud triggers are observable in real time.

Level 5 — AI-Ready, Regulator-Ready Architecture

Every decision path is transparent, auditable, explainable, and aligned with supervisory expectations.

PTMM gives banks a measurable way to evaluate modernization progress and identify architectural gaps that must be addressed before deploying AI into payment flows.

Regulatory pressure on payment transparency

Payment systems are increasingly treated as **national infrastructure**, and regulators are shifting toward architecture-level expectations. Supervisory bodies now emphasize:



- Real-time fraud explainability
- Traceability of payment routing decisions
- Auditability of automated outcomes
- ISO 20022 semantic consistency
- Operational resilience in real-time payment rails
- Governance of AI-driven payment decisions

As real-time payment schemes (FedNow, RTP, UPI, SEPA Instant) expand, regulators will require banks to demonstrate not only how AI models work, but how the **payment architecture supports transparent, safe, and explainable decisioning**.

Legacy systems cannot meet these expectations without modernization.

A blueprint for AI-ready payment modernization

Banks preparing for AI-enabled payment operations should adopt a structured blueprint:

- Establish real-time data lineage across all payment rails
- Decouple routing and exception logic from monolithic cores
- Introduce architectural observability for fraud, STP, and exception flows
- Implement governance layers that track model inputs, outputs, and overrides
- Build modernization roadmaps that prioritize transparency, not speed
- Align payment modernization with regulatory expectations for explainability

This blueprint ensures AI is deployed into payment environments where decisions are explainable, risks are manageable, and operations are resilient.

Why this matters for the future of payments

Payments are the heartbeat of the financial system. As banks adopt AI to improve fraud detection, routing, and STP, the underlying architecture must evolve to support transparency and explainability.

Institutions that adopt transparency-first payment modernization will be the ones capable of:

- Meeting emerging regulatory expectations
- Deploying AI safely and responsibly
- Reducing fraud and operational risk
- Supporting real-time payment schemes
- Modernizing without destabilizing legacy cores

Architectural transparency is no longer optional.

It is the foundation of AI-ready payments.



About the Author

Neeraj Aggarwal is a modernization and payments transformation leader with deep expertise in AI-enabled banking, core modernization, and real-time payment architectures. He advises financial institutions on building resilient, transparent, and regulator-ready payment systems and contributes thought leadership across global industry forums.

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Regulators are already asking for explainability in AI-driven decisioning, but payments remain the least transparent part of the bank. Legacy routing logic, undocumented exception flows, and batch-era data paths make it impossible to justify real-time decisions with confidence. The industry keeps investing in AI c



the surface while ignoring the architectural debt underneath. Without modernizing the payment core for traceability and real-time observability, AI adoption will continue to create more risk than value.

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This resonates with what many institutions discover once they begin serious modernization work. Payment engines built decades ago simply weren't designed for real-time observability or explainable decision flows. AI exposes every hidden dependency and undocumented rule. The shift toward AI-ready payments isn't about adding intelligence—it's about restructuring the core so intelligence can operate safely and predictably.

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